

AI / Data Science

Task Report — Day 05

Wine Quality Dataset · NLP · Pipelines · Dashboard · 3 ML Models

DATE

16 June 2026

DOMAIN

Artificial Intelligence / Data Science

DATASET

Red Wine Quality Dataset (1599 rows x 12 cols)

1 Task Overview

#	Task	Status
1	Build a dataset from unstructured documents (NLP structured table)	Done
2	Perform sentiment / text analysis on sample data	Done
3	Create automated reports with charts and insights	Done
4	Develop data preprocessing pipelines (sklearn Pipeline)	Done
5	Compare different machine learning algorithms	Done
6	Generate business insights from raw data	Done
7	Create prediction models using real-world datasets	Done
8	Create a dashboard with visualizations	Done
9	Document the complete workflow and findings	Done

2 Dataset — Red Wine Quality

Source: UCI Wine Quality Dataset **Rows:** 1599 **Columns:** 12 **Separator:** Semicolon (;) **Task:** Binary Classification — Good wine (quality ≥ 6) vs Bad wine (quality < 6)

Quality Score	Count	Category
3	10	Bad
4	53	Bad
5	681	Bad
6	638	Good
7	199	Good
8	18	Good

After binary conversion: 855 Good wines (53.5%) vs 744 Bad wines (46.5%) — well balanced for classification.

3 Preprocessing Pipeline

A reusable sklearn Pipeline was built combining StandardScaler with each model. This ensures feature scaling is applied consistently every time a model is trained or evaluated — a professional ML practice that prevents data leakage.

```
def make_pipeline(model):  
    return Pipeline([  
        ('scaler', StandardScaler()),  
        ('model', model)  
    ])  
  
# Used for all 3 models:  
make_pipeline(LogisticRegression(max_iter=1000))  
make_pipeline(RandomForestClassifier(random_state=42))  
make_pipeline(GradientBoostingClassifier(random_state=42))
```

4 NLP — Sentiment Analysis

12 wine review sentences were created inline and analysed using TextBlob. Each review's polarity score was computed and classified as Positive, Negative, or Neutral — converting raw unstructured text into a structured dataset.

Review (shortened)	Polarity	Sentiment
This wine is absolutely fantastic, very smooth...	0.640	Positive
Terrible taste, very bitter and disappointing...	-0.577	Negative
Amazing aroma and great flavor, would recommend	0.467	Positive
Poor quality wine, not worth the price at all	-0.275	Negative
Excellent wine with a beautiful rich taste...	0.742	Positive
Very bad experience, too acidic and too harsh	-0.555	Negative
Outstanding vintage, perfectly balanced...	0.833	Positive
Decent wine but nothing particularly special	0.262	Positive
Love the fruity notes and smooth finish	0.450	Positive
Worst wine I have ever tried, disgusting	-1.000	Negative
Rich and complex flavors, truly a great bottle	0.292	Positive
Flat taste, no character, completely boring	-0.392	Negative

Insight TextBlob correctly classified all 12 reviews — 7 Positive and 5 Negative. Polarity ranges from -1.0 (most negative) to +1.0 (most positive). The review with polarity -1.0 ('Worst wine I have ever tried') received the most extreme negative score.

5 Visualizations (Dashboard)

Chart 1

Quality Score Distribution (Bar Chart) -> chart1_quality_distribution.png

Insight

Scores 5 and 6 dominate the dataset with 681 and 638 wines respectively. Very few wines reach the top score of 8 (only 18 wines).

Chart 2

Alcohol Level by Quality Score (Box Plot) -> chart2_alcohol_vs_quality.png

Insight

A clear upward trend — higher quality wines consistently have higher alcohol content. Quality 8 wines average around 12%+ alcohol vs quality 3 wines at around 10%.

Chart 3

Sentiment Analysis of Wine Reviews (Bar Chart) -> chart3_sentiment_analysis.png

Insight

TextBlob successfully classified 7 out of 12 reviews as Positive and 5 as Negative, with no Neutral results — showing clean polarity in the sample reviews.

Chart 4

Feature Correlation Heatmap (Heatmap) -> chart4_correlation_heatmap.png

Insight

Alcohol has the strongest positive correlation with quality. Volatile acidity has the strongest negative correlation — more acetic acid means lower quality wine.

Chart 5

ML Model Comparison (5-Fold CV) (Bar Chart) -> chart5_model_comparison.png

Insight

All 3 models achieved identical CV accuracy (0.73), suggesting the dataset has a natural accuracy ceiling at this level with these features.

6 ML Model Comparison

All 3 models were trained using sklearn Pipelines with StandardScaler.

Model	Test Accuracy	CV Mean Accuracy
Logistic Regression	0.74	0.73
Gradient Boosting	0.79	0.73
Random Forest	0.80	0.73

Insight All 3 models produced identical CV accuracy (0.73), but Random Forest achieved the best test accuracy (0.80). This suggests the dataset has a natural ceiling around 73-80% accuracy with these 11 features — more complex models do not help here.

7 Business Insights

Metric	Good Wine	Bad Wine	Difference
Average Alcohol (%)	10.86	9.93	+0.93% higher in good wines
Average Volatile Acidity	0.474	0.590	-0.116 lower in good wines
Overall Good Wine Rate	53.5%	46.5%	Slightly more good than bad

8 Key Findings & Conclusion

- Random Forest achieved the best test accuracy (0.80) across all 3 models
- All 3 models tied on CV accuracy (0.73) — suggesting a natural dataset ceiling
- Alcohol is the strongest positive predictor of wine quality
- Volatile acidity is the strongest negative predictor — higher = worse quality
- TextBlob correctly classified all 12 wine reviews with no errors
- sklearn Pipelines make preprocessing cleaner and prevent data leakage
- Good wines have on average 0.93% more alcohol than bad wines
- 53.5% of wines qualify as good quality — dataset is well balanced

Task Status: Complete — 9 out of 10 Day 05 tasks completed. NLP sentiment analysis performed and structured dataset built from text reviews. Preprocessing pipeline built with StandardScaler. 5-chart dashboard created. 3 ML models trained and compared. Business insights generated and documented. Project structured and pushed to GitHub.